

รายงานฉบับสมบูรณ์โครงการรวบยอดวิศวกรรมคอมพิวเตอร์ 2

CEDT Capstone Project Final Report

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Title in English

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บทคัดย่อ

โครงการนี้มีชื่อว่า [ชื่อโครงการภาษาไทย] มุ่งเน้นการแก้ปัญหา [ระบุปัญหาหลัก] ซึ่งมีความสำคัญเนื่องจาก [แรงจูงใจ/ความสำคัญในโลกจริง] วัตถุประสงค์ของโครงการคือ [ระบุวัตถุประสงค์หลัก 2-3 ข้อ]

แนวทางการแก้ปัญหาประกอบด้วย [สรุปวิธีการ/แนวคิดการออกแบบ] โดยผลงาน [องค์ประกอบฮาร์ดแวร์/ซอฟต์แวร์/ฐานข้อมูล/เครือข่าย] ระบบถูกพัฒนาโดยใช้ [เครื่องมือหรือแพลตฟอร์มหลัก] และผ่านการทดสอบด้วย [ประเภทของการทดลองหรือการประเมินผล]

ผลการดำเนินงานแสดงให้เห็นว่า [สรุปผลลัพธ์สำคัญ] โดยบรรลุ [เน้นค่าประสิทธิภาพ/ความถูกต้อง/ประโยชน์ที่สำคัญ] โครงการนี้มีส่วนช่วยต่อ [ผลกระทบเชิงวิชาการ/อุตสาหกรรม/สังคม] ผ่าน [สรุปการมีส่วนร่วม]

คำสำคัญ: [คำสำคัญ 1], [คำสำคัญ 2], [คำสำคัญ 3], [คำสำคัญ 4], [คำสำคัญ 5]

Abstract

This project, titled [Project Title], addresses the problem of [state the main problem], which is significant due to [motivation/real-world importance]. The objectives of this project are [list 2-3 main objectives].

The proposed solution involves [summarize the methodology/design concept], integrating [hardware/software/database/networking elements]. The system was implemented using [main tools, frameworks, or platforms] and evaluated through [type of experiments/tests].

Results indicate that the system [summarize major findings], achieving [highlight key performance/accuracy/benefits]. The project contributes to [academic/industrial/societal impact] by [state contributions].

Keywords: [Keyword 1], [Keyword 2], [Keyword 3], [Keyword 4], [Keyword 5]

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1 Introduction & Stakeholder Context

1.1 Background & Motivation

Why this problem matters in professional/real-world settings.

1.2 Problem Statement

What problem was addressed, and why is it complex?

1.3 Objectives

Final goals pursued.

1.4 Scope & Limitations

Boundaries of the project.

1.5 Stakeholders

Who benefits (end-users, organizations, community).

1.6 Expected Benefits

Technical, societal, industrial, or academic impact.

2 Related Theory and Research

2.1 Review of Existing Knowledge

Prior work, models, or systems related to the project.

2.2 Comparison with Existing Systems

Identify strengths/weaknesses.

2.3 Gap Linkage to Project

Identify research/technology gaps filled by this project.

3 System Design

3.1 System Architecture

Block diagrams, workflows, UML diagrams.

3.2 Components

Hardware, software, databases, networking.

3.3 Design Considerations

Performance, scalability, usability, sustainability, environment.

3.4 Proposed vs. Final Design

Adjustments from proposal to implementation.

4 Implementation

4.1 Development Process

Modules, algorithms, prototypes built.

4.2 Tools, Frameworks, or Environments

Programming languages, libraries, development platforms, testing environments.

4.3 Challenges Encountered and Resolutions

Technical, design, or integration difficulties and how the team overcame them.

4.4 Evidence of Functional Prototype/System

Screenshots, architecture logs, photos, demo references.

5 Testing and Experimentation

5.1 Experiment/Test Plan

What was tested and why.

5.2 Setup

Hardware/software environment, datasets, conditions.

5.3 Metrics

Accuracy, efficiency, usability, sustainability, etc.

5.4 Test Cases

Key scenarios for validation.

6 Results and Analysis

6.1 Results Presentation

Summarize findings clearly with tables, charts, and graphs. Evaluate outcomes against stated objectives, benchmarks, or standards.

6.2 Analysis & Interpretation

Explain patterns, significance, and implications of results.

6.3 Limitations & Lessons

Discuss unexpected issues, anomalies, and what was learned.

7 Project Management and Teamwork

7.1 Timeline & Milestones

Show progress made and key deliverables completed.

7.2 Work Division

Describe task allocation and balance among members.

7.3 Communication

Explain tools, meetings, and methods for collaboration.

7.4 Reflection

Assess strengths, weaknesses, and improvements in teamwork.

8 Ethics, Privacy, and Professional Considerations

8.1 Ethical Issues

Describe how privacy, fairness, safety, and inclusivity are handled.

8.2 Standards & Laws

Identify codes of conduct, legal requirements, or regulations applied.

8.3 Environmental Impact

Note measures for efficiency, sustainability, or reduced footprint.

8.4 Intellectual Property

Clarify ownership, licensing, or agreements with partners.

9 Risk and Resource Management

9.1 Risks

Identified risks and mitigation strategies.

9.2 Adaptations

Adjustments during project development.

9.3 Resources

Hardware, software, datasets, financial usage.

9.4 Cost Considerations

Rough budgeting or cost estimation.

10 Conclusion and Future Work

10.1 Achievements

Summarize main results, system features, and contributions.

10.2 Objectives Met

Evaluate how well the project goals were fulfilled.

10.3 Applications

Point out technical, industrial, or societal uses.

10.4 Future Work

Suggest next steps, enhancements, or research directions.

References

1. A. Author, B. Author, and C. Author, “Title of the article,” *Journal Name*, vol. 10, no. 2, pp. 123–135, Feb. 2021.
2. D. Author, *Book Title*. 2nd ed. New York, NY, USA: Publisher, 2020.
3. E. Author, “Title of Conference Paper,” in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, Seoul, Korea, 2019, pp. 456–462.
4. F. Author, “Title of Webpage,” Website Name, 2022. [Online]. Available: <https://www.example.com> [Accessed: Sep. 9, 2025].

A Source Code Excerpts

Provide relevant code snippets (not the entire codebase). Highlight key functions or algorithms. Add comments/explanations where necessary.

B Circuit Diagrams / Schematics

Include electronic or hardware diagrams. Label all components clearly. Provide captions and references.

C User Interface Mockups / Screenshots

Show UI designs, mockups, or final application screenshots. Include captions and references to sections in the main report.

D Extended Test Data

Detailed logs, datasets, or evaluation results. Tables or charts that support the analysis in the main body. Notes on data collection methods (if applicable).

E Additional Documentation

User manuals or setup instructions. Consent forms (if user study is involved). Contracts or agreements with collaborators/companies.